Reagan Chiles

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Education

University of Colorado Boulder 2024–2028
B.S. Mechanical Engineering GPA: 3.51

Experience

J.S. Held May 2025–August 2025
Mechanical Engineering Intern

- Applied engineering and forensic principles to reconstruct vehicle crashes for major manufacturers.
- Used 3D scanners to collect, analyze, and process point-cloud data.
- Collected and analyzed data and tested mechanical systems to support technical reports.
- Organized and filed technical data and documentation for depositions and trials.

Speedy Sparkle Car Wash August 2025–Present
Supervisor

- Operate, maintain, and repair mechanical wash assemblies.
- Manage teams to complete tasks efficiently.
- Apply customer service and attention to detail in daily operations.

Projects

1U CubeSat Structural Design and MBSE *Personal Engineering Project*

- Developed a 1U CubeSat structural design using model-based systems engineering (MBSE) and CATIA Magic SysML v2.
- Established engineering requirements using satisfy and verify relationships to link requirements to design elements and verification cases.
- Developed a SolidWorks CAD model of a PETG CubeSat with system level requirements for FDM manufacturing and verification testing

Custom 3D-Printed Medical Brace *Personal Design Project*

- Designed a lightweight wrist/forearm brace using 3D-scanned geometry and CAD for a personalized fit.
- Developed an open lattice structure to reduce material usage by 50% while maintaining structural support.
- Manufactured and iterated FDM prototypes, evaluating fit, geometry, and manufacturability.

Ratchet Strap Assembly FEA *SolidWorks Simulation*

- Reverse-engineered a commercial ratchet strap assembly through physical measurements and analysis.
- Developed a detailed SolidWorks CAD model, recreating component geometry and assembly relationships.
- Performed static FEA using defined materials, constraints, loads, and mesh parameters to evaluate stress, deformation, and potential failure regions.

Engineering Involvement

Formula SAE *Member*

- Contribute to vehicle dynamics and chassis design for a Formula SAE race car.

American Institute of Aeronautics and Astronautics (AIAA) *Member*

Technical Skills

CAD:	SolidWorks (CSWA), 3D CAD modeling, FEA, CFD
Systems Engineering:	CATIA Magic, MBSE, requirements development, verification planning
Programming:	MATLAB, C (50+ hours)
Forensics:	Vehicle dynamics, 3D scanning, point-cloud analysis,
Manufacturing:	FDM and SLA 3D printing, MIG welding, lathing, milling,design for manufacturing
Other Software:	Microsoft Excel and Microsoft Word (100+ hours)